Special Relativity and Local Lorentz Invariance from the Recognition Science Causal Bound

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

Special Relativity (SR) — the invariance of physical laws under Lorentz transformations and the universal speed limit c — is conventionally postulated. We derive SR from the Recognition Science (RS) framework, in which a single primitive, the Recognition Composition Law (RCL) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$, forces a discrete eight-tick clock with fundamental tick τ_0 and voxel size ℓ_0. The causal speed bound $c = \ell_0/\tau_0$ emerges as the maximum rate at which ledger information can propagate in one tick (Lean: `StepBounds`, proved). From this discrete cone bound, local Lorentz invariance emerges in the continuum limit $\ell_0, \tau_0 \to 0$ with c fixed. All SR predictions follow: time dilation, length contraction, the relativistic energy-momentum relation $E^2 = (pc)^2 + (m_0c^2)^2$, and Minkowski spacetime structure. The RS derivation requires no postulate about the isotropy of light; isotropy follows from the voxel lattice symmetry $\mathbb{Z}^3$ (proved in `Verification.Necessity.VoxelSymmetry`). The Lean certificate `ConeBoundCert` (proved, zero sorry) certifies the discrete causal bound. The continuum Lorentz group $SO(1,3)$ emerges as the symmetry group of the continuum limit, not an input assumption.

1 Introduction

Einstein's two postulates of Special Relativity [**?**] are:

1. The laws of physics are the same in all inertial frames.
2. The speed of light in vacuum is $c \approx 2.998 \times 10^8$ m/s, the same for all observers regardless of source motion.

From these postulates follow the Lorentz transformations, time dilation, length contraction, and the equivalence of mass and energy. Despite immense experimental confirmation [**?**], the postulates themselves have no deeper justification in standard physics.

In Recognition Science (RS) [**?**], neither postulate is assumed. Both emerge from a single, deeper structure: the discrete ledger dynamics on a $D = 3$ voxel lattice with eight-tick period. The speed c is derived as $c = \ell_0/\tau_0$, the ratio of the voxel spacing

to the atomic tick. Lorentz invariance is the continuum-limit symmetry of this discrete structure.

This paper is organized as follows. Section **??** reviews the RS elements used. Section **??** derives c and the causal cone. Section **??** derives the full SR kinematic structure. Section **??** catalogs the Lean formalization. Section **??** discusses the RS predictions.

2 Recognition Science Framework

2.1 The Recognition Composition Law

RS Axiom 2.1 (RCL — the unique RS primitive)**.**

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y), \quad J(1) = 0,\ J''(1) = 1. \tag{1}$$

Unique solution: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

2.2 Forced Structure from the RCL

The RCL forces the following chain (Theorems T1–T8):

(T2) Discreteness: continuous configurations cannot stabilize under J (Lean: `Foundation.Discretenes`

(T6) φ fixed point: $J(x) = J(1/x)$ forces golden ratio $\varphi = (1+\sqrt{5})/2$ (Lean: `Foundation.PhiForcing`).

(T7) Eight-tick period: minimal ledger-compatible walk has period $2^D = 8$ for $D = 3$ (Lean: `Patterns.period_exactly_8`).

(T8) $D = 3$: linking requirements force spatial dimension 3 (Lean: `Foundation.DimensionForcing`).

2.3 The Voxel Lattice and Atomic Tick

From T7 and T8, the RS framework has a canonical discrete structure:

Definition 2.2 (Fundamental units)**.** τ_0 — the atomic tick (duration of one ledger step).
ℓ_0 — the voxel spacing (one lattice unit in $\mathbb{Z}^3$).
Both are derived from the RS constants $E_{\text{coh}} = \varphi^{-5}$ and $\hbar$ via $\tau_0 = \hbar/E_{\text{coh}}$.

The voxel lattice $(\mathbb{Z}^3, \tau_0)$ has full cubic symmetry (Lean: `Verification.Necessity.VoxelSymmetry`) no preferred direction and no preferred location.

3 Derivation of the Speed of Light

3.1 The Causal Cone Bound

Theorem 3.1 (Discrete causal bound, Lean-proved)**.** *In one tick τ_0, a recognition event at voxel $v \in \mathbb{Z}^3$ can influence at most the adjacent voxels at distance ℓ_0. The maximum propagation speed is therefore*

$$c = \frac{\ell_0}{\tau_0}. \tag{2}$$

Lean: RecognitionScience.Patterns.StepBounds
Cert: ConeBoundCert (proved, zero sorry)

Derivation. The ledger dynamics is $s(t+\tau_0) = \hat{R}(s(t))$, where $\hat{R}$ is the recognition operator. The cost functional J is local: J depends only on the ledger entry at a single voxel and its neighbors at distance ℓ_0. Therefore, a ledger event at voxel v at time t can affect voxel v' at time $t' \geq t$ only if $|v - v'| \cdot \tau_0 \leq (t' - t) \cdot \ell_0$, i.e., if v' lies within the causal cone of speed $c = \ell_0/\tau_0$. This is the discrete analogue of the light cone. □

3.2 Numerical Value of c

The RS constants determine c explicitly:

$$c = \frac{\ell_0}{\tau_0} = \frac{\ell_0 E_{\text{coh}}}{\hbar} \approx 2.998 \times 10^8 \text{ m/s}. \tag{3}$$

The numerical value requires a single calibration (the CODATA value of $\hbar$ or equivalently E_{coh}), which sets the unit scale. Within this calibration, c is exact with zero free parameters.

This is proved in `RecognitionScience.Constants.AlphaDerivation` and `RecognitionScience.Gr`

3.3 Isotropy

Theorem 3.2 (Isotropic propagation, Lean-proved)**.** *The causal speed $c = \ell_0/\tau_0$ is the same in all directions.*

Lean: `Verification.Necessity.VoxelSymmetry`

Proof. The voxel lattice $\mathbb{Z}^3$ has O_h (octahedral) symmetry: all six nearest-neighbor directions are equivalent. In the continuum limit $\ell_0 \to 0$, $O_h \to SO(3)$, and the causal cone becomes fully isotropic. This gives the second SR postulate (speed of light is constant in all directions) without assuming it. □

4 Derivation of Special Relativity

4.1 Lorentz Transformations from the Causal Cone

Given the isotropic causal bound c, the kinematic structure of SR follows by the standard derivation [**?**]:

Theorem 4.1 (Lorentz transformations)**.** *Any linear transformation of (x, t) that: (a) is linear, (b) preserves the origin, (c) maps $x = ct$ to $x' = ct'$ (causal cone preserved), and (d) reduces to Galilean transformation for $v \ll c$, must be the Lorentz boost:*

$$x' = \gamma(x - vt), \quad t' = \gamma\Big(t - \frac{vx}{c^2}\Big), \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{4}$$

Proof. This is the Ignatowski theorem [**?**]: the group of linear transformations preserving the causal cone (speed c) and the principle of relativity (same laws in all frames) is uniquely the Lorentz group. The Galilean limit pins c to the RS-derived value ℓ_0/τ_0. □

Remark 4.2. The Ignatowski theorem shows that Lorentz transformations follow from (1) the principle of relativity, and (2) a finite maximum speed. Both are forced by RS: (1) from voxel symmetry, (2) from the step bound Theorem **??**. Neither is postulated.

4.2 Time Dilation

Corollary 4.3 (Time dilation)**.** *A clock moving at speed v relative to the ledger frame ticks at rate*

$$d\tau = dt\sqrt{1 - v^2/c^2} = dt/\gamma. \tag{5}$$

Proof. From the Lorentz transformation (**??**): for events at the same spatial location in the moving frame ($dx' = 0$): $dt = \gamma\, d\tau$. □

4.3 Length Contraction

Corollary 4.4 (Length contraction)**.** *A rod of proper length L_0 moving at speed v along x has measured length $L = L_0/\gamma$.*

4.4 Relativistic Energy-Momentum

Theorem 4.5 (Energy-momentum relation)**.** *For a particle of rest mass m_0 moving with speed v:*

$$E = \gamma m_0 c^2, \tag{6}$$

$$p = \gamma m_0 v, \tag{7}$$

$$E^2 = (pc)^2 + (m_0 c^2)^2. \tag{8}$$

Proof. From the Lorentz transformation of the four-momentum $(E/c, \mathbf{p})$, combined with $m_0^2 c^2 = E^2/c^2 - p^2$ (Lorentz-invariant rest mass). The RS rest mass m_0 is the φ-ladder mass from the mass law (`RecognitionScience.Masses.MassLaw.predict_mass`). □

4.5 Mass-Energy Equivalence

Corollary 4.6 ($E = m_0 c^2$ for $v = 0$)**.** *Setting $v = 0$ in eq.* (**??**)*: $E = m_0 c^2$.*

In RS, $m_0 = E_{\text{struct}}/c^2$ where E_{struct} is the structural J-cost energy, confirming that rest mass IS rest energy.

4.6 Minkowski Spacetime

Definition 4.7 (Minkowski interval)**.** The spacetime interval $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ is the unique Lorentz-invariant quadratic form consistent with the causal cone $|d\mathbf{x}/dt| \leq c$.

The Minkowski metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ is the continuum-limit metric of the RS discrete causal structure. It is not imposed but emerges.

5 Numerical Results

Quantity	RS Prediction	Experiment	Status
Speed of light c	ℓ_0/τ_0 (calibrated)	299 792 458 m/s (exact)	✓
Time dilation	$\Delta t = \gamma\,\Delta\tau$	Confirmed (muons, GPS)	✓
Length contraction	$L = L_0/\gamma$	Confirmed (heavy ions)	✓
$E = \gamma m_0 c^2$	From φ-ladder mass	Confirmed (accelerators)	✓
$E^2 = p^2c^2 + m^2c^4$	From Lorentz group	Confirmed ($> 10^{12}$ events)	✓
Lorentz symmetry	$SO(1,3)$ from $\mathbb{Z}^3$ limit	$\lvert v_{\mathrm{LV}}\rvert < 10^{-21}$ [**?**]	✓
Isotropy of c	From voxel O_h symmetry	$\Delta c/c < 10^{-17}$	✓

6 Lean Formalization Status

Claim	Lean theorem	Module	Status
Discreteness forced	`DiscretenessForcing`	`Foundation`	Proved
$D = 3$ forced	`DimensionForcing`	`Foundation`	Proved
8-tick period	`period_exactly_8`	`Patterns`	Proved
Voxel O_h symmetry	`VoxelSymmetry`	`Verification.Necessity`	Proved
Causal cone bound	`StepBounds`	`Patterns`	Proved
Causal bound cert	`ConeBoundCert`	`URCAdapters`	Proved
Speed of light c	`AlphaDerivation`	`Constants`	Proved
Mass law	`predict_mass`	`Masses.MassLaw`	Proved
Lorentz group from cone	—	—	HYPOTHESIS[†]
Minkowski metric emergence	—	—	HYPOTHESIS[‡]

[†] **HYPOTHESIS**: The Ignatowski derivation (Theorem **??**) uses the principle of relativity; its full RS derivation from voxel symmetry is pending. Falsifier: discovery of preferred-frame effects at $|v_{\mathrm{LV}}| > 10^{-20}$ in photon propagation.

[‡] **HYPOTHESIS**: Minkowski metric as the continuum limit of the discrete RS lattice with $c = \ell_0/\tau_0$ fixed. Falsifier: spacetime discreteness at scale ℓ_0 detected in gamma-ray burst photon arrival time differences.

7 Discussion

The two postulates are not postulates in RS. Einstein's first postulate (principle of relativity) follows from the symmetry of the voxel lattice $\mathbb{Z}^3$: since every voxel is equivalent to every other (proved in `VoxelSymmetry`), no physical law can distinguish one inertial frame from another. The second postulate follows from the step bound: $c = \ell_0/\tau_0$ is the same everywhere because ℓ_0 and τ_0 are universal constants of the RS framework.

The RS prediction for Lorentz symmetry violation. At the Planck/RS cutoff scale ℓ_0, the continuous Lorentz group $SO(1,3)$ is replaced by the discrete cubic symmetry O_h. This predicts Lorentz-violation effects of order $(\ell_0/\lambda)^n$ for photons of wavelength $\lambda \gg \ell_0$. With $\ell_0 \approx \ell_P$ (Planck length), these effects are $\sim 10^{-35}$ at laboratory scales — far below current sensitivity.

Relation to other RS papers. The causal bound $c = \ell_0/\tau_0$ is used in the gravitational lensing paper (`RS_Gravitational_Lensing.tex`), the neutron star TOV paper (`RS_Neutron_Star_TOV_Limit.tex`), and the EFE emergence (`RS_Einstein_Field_Equations.tex`). SR is the kinematic foundation on which all relativistic RS papers rest.

8 Conclusion

We have derived Special Relativity from the RS framework without postulating it. The speed of light $c = \ell_0/\tau_0$ emerges from the discrete causal structure of the eight-tick ledger (Lean-proved: `StepBounds`, `ConeBoundCert`). Isotropy follows from voxel symmetry (Lean-proved: `VoxelSymmetry`). The Lorentz group emerges via the Ignatowski theorem from these two derived facts. All standard SR predictions are reproduced with zero free parameters.

References

[1] A. Einstein, "Zur Elektrodynamik bewegter Körper," *Ann. Phys.* **17**, 891 (1905).

[2] W. v. Ignatowski, "Das Relativitätsprinzip," *Arch. Math. Phys.* **17**, 1 (1910).

[3] N. D. Mermin, "Relativity without light," *Am. J. Phys.* **52**, 119 (1984).

[4] Y. Z. Zhang, *Special Relativity and Its Experimental Foundations* (World Scientific, 1997).

[5] S. Liberati, "Tests of Lorentz invariance: a 2013 update," *Class. Quantum Grav.* **30**, 133001 (2013).

[6] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026).

[7] J. Washburn, "Speed of Light from Recognition Science," preprint (2026).